Advanced Programming Practice · 21CSC203P

Unit 5 · Session 5 · SLO 2 — Solved Worksheet

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Unit â€“ 5 : Session â€“ 5 : SLO â€“ 2 | Polynomial and Rational Expression Operations

I. Questions & Answers

1. Multiply (x+2)(x+3) â€” expand((x + 2)*(x + 3)) â†’ x**2 + 5*x + 6.

2. Factor xÂ² â^’ 4 â€” factor(x**2 â^’ 4) â†’ (x â^’ 2)*(x + 2) (difference of squares).

3. Add 1/(x+1) + 1/(xâ^’1) â€” together/simplify: combine over the common denominator (x+1)(xâ^’1) â†’ ((x â^’ 1) + (x + 1)) / (xÂ² â^’ 1) = 2*x/(x**2 â^’ 1).

II. Code

- from sympy import symbols, expand, factor, simplify
- x = symbols('x')
- print(expand((x + 2)*(x + 3))) # x**2 + 5*x + 6
- print(factor(x**2 - 4)) # (x - 2)*(x + 2)
- print(simplify(1/(x + 1) + 1/(x - 1))) # 2*x/(x**2 - 1)